PROJECT - 4

Deploying a Web Application into Different Environments using GitHub Actions

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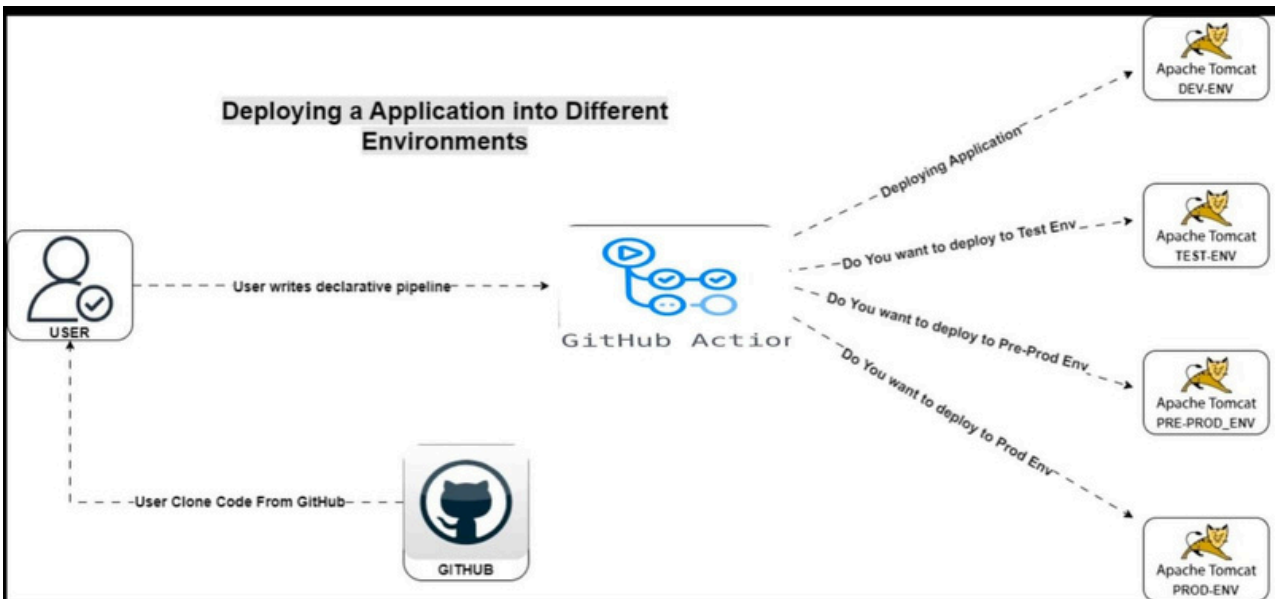


High-level architecture showing how GitHub Actions deploys the application into different environments (DEV, TEST, PRE-PROD, and PROD)

Project Overview

This project demonstrates how to automate the deployment of a web application to different environments (DEV, TEST, PRE-PROD, PROD) using GitHub Actions. The goal is to achieve smooth and reliable CI/CD pipeline automation integrated with Apache Tomcat servers.

Objective

- Automate deployment using GitHub Actions.
- Deploy to multiple environments based on user input.
- Reduce manual errors and make deployments faster.

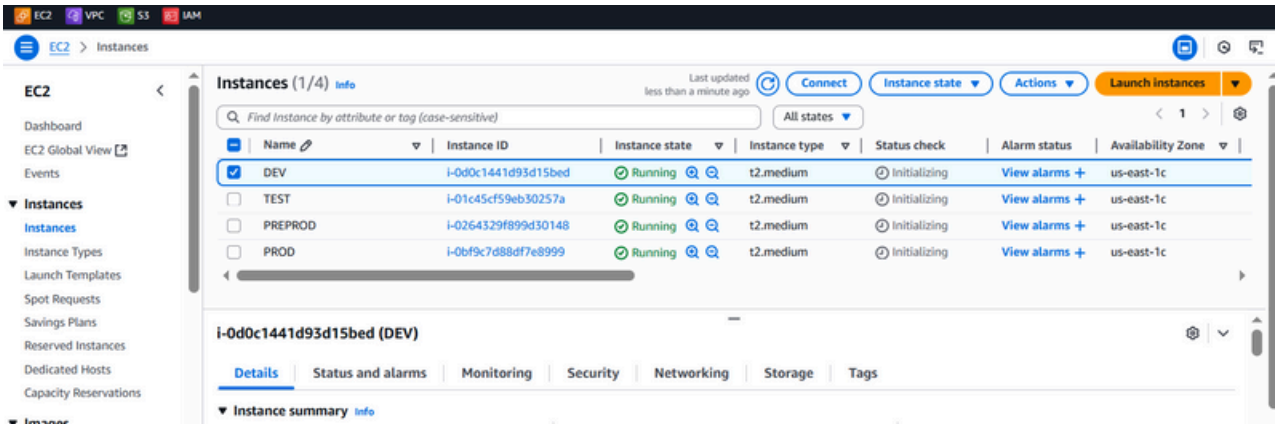
Tools and Technologies Used

- **Version Control:** Git, GitHub
- **CI/CD Tool:** GitHub Actions
- **Web Server:** Apache Tomcat
- **Build Tool:** Maven
- **Cloud Platform:** AWS EC2 (for hosting Tomcat)
- **Languages:** YAML, Java, HTML

AWS EC2 Server Setup

The project uses four EC2 instances to host Apache Tomcat for different environments:

- DEV – Development server
- TEST – Testing server
- PRE-PROD – Pre-Production server
- PROD – Production server

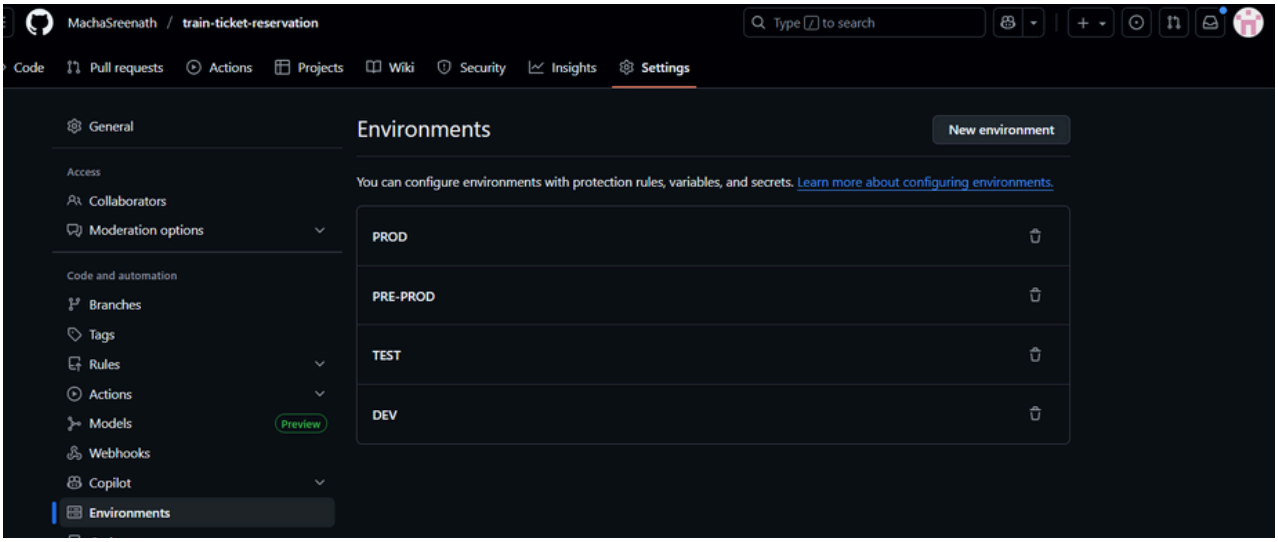


Tomcat was installed on the DEV server, and an AMI was created from it to quickly set up TEST, PRE-PROD, and PROD servers.

GitHub Actions Environment Setup and Deployment

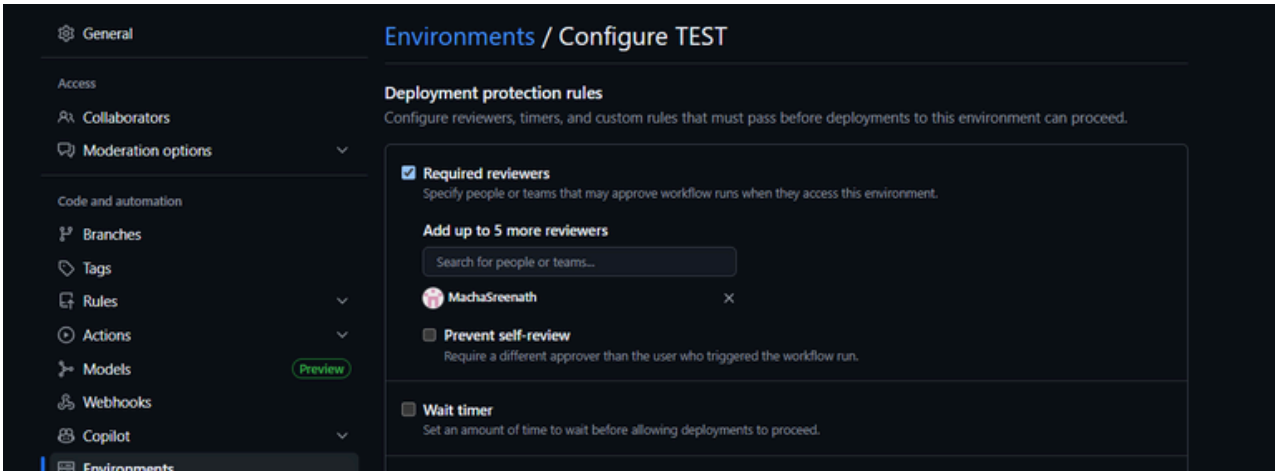
1. Creating Environments in GitHub Actions

   - Four environments were created: DEV, TEST, PRE-PROD, and PROD.
   - Each environment corresponds to a Tomcat server in AWS EC2.



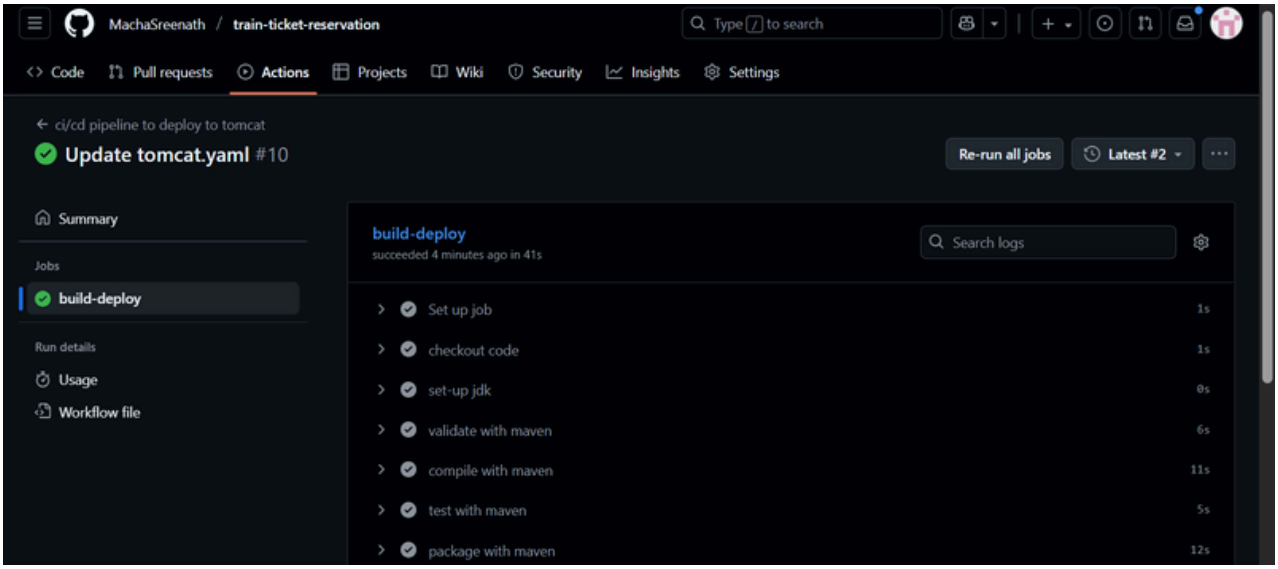
2. Enabling Manual Approval

   - **DEV environment:** Deployment runs automatically without approval.
   - **TEST, PRE-PROD, and PROD environments:** Deployment requires at least one person's manual approval in GitHub Actions before proceeding to the next environment.



3. Pipeline Execution

- After committing code, the pipeline ran successfully.
- DEV, TEST, and PRE-PROD deployments were executed automatically, and PROD deployment waited for approval.



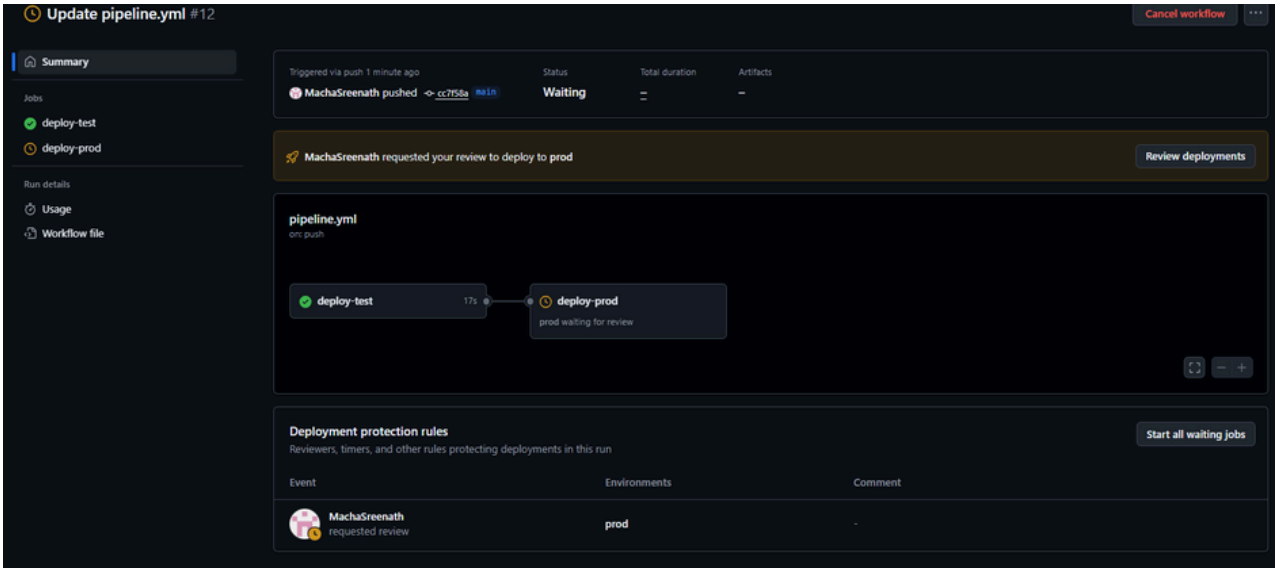
GitHub Actions pipeline run success.

Manual Approval for Deployment

The workflow is configured so that deployment to the next environment requires manual approval.

- DEV environment: Runs automatically.
- TEST, PRE-PROD, and PROD environments: The pipeline pauses and waits for at least one person's approval before continuing.

Note: The image below shows an example of the approval request. This is similar to what appears in the TEST, PRE-PROD, and PROD environments when manual approval is required.

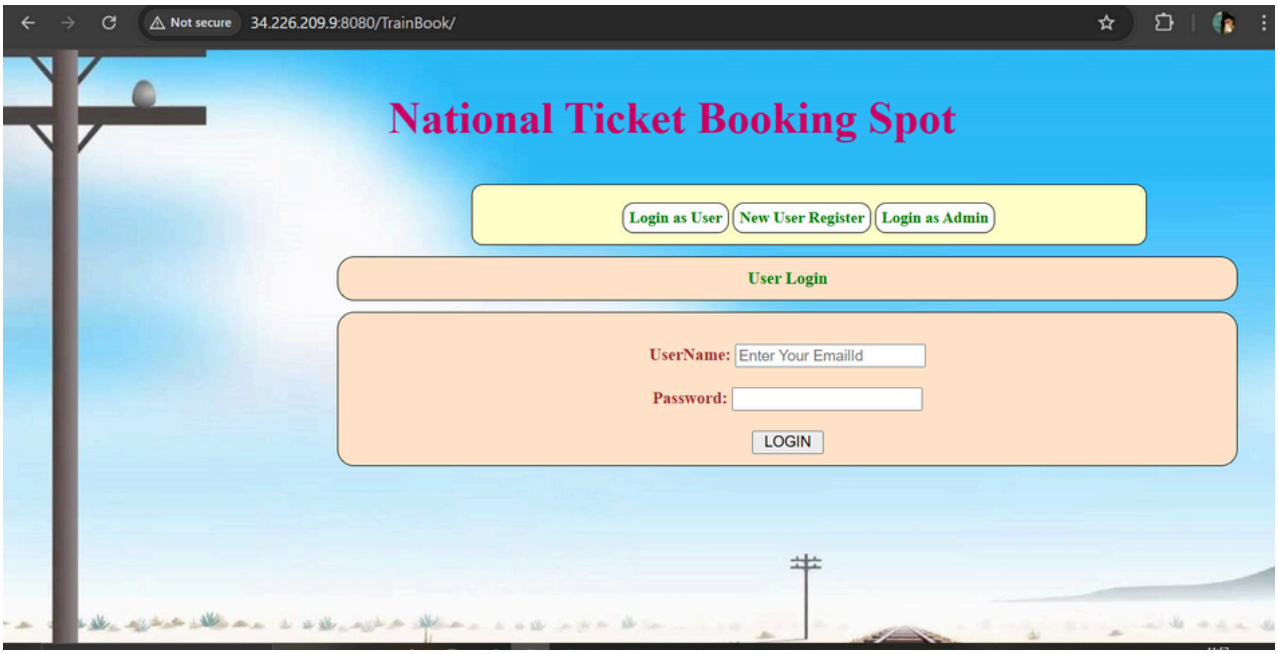


Example of manual approval request in GitHub Actions.

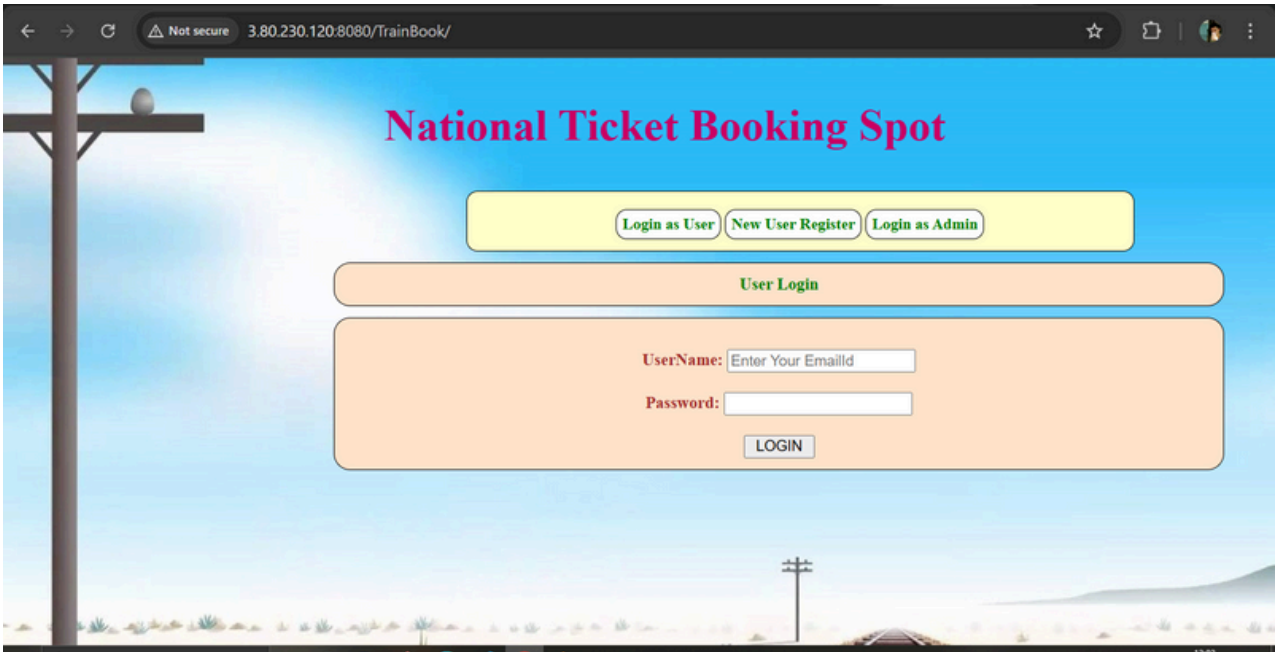
Application Deployment on Servers

The Java web application was successfully deployed to all four environments.

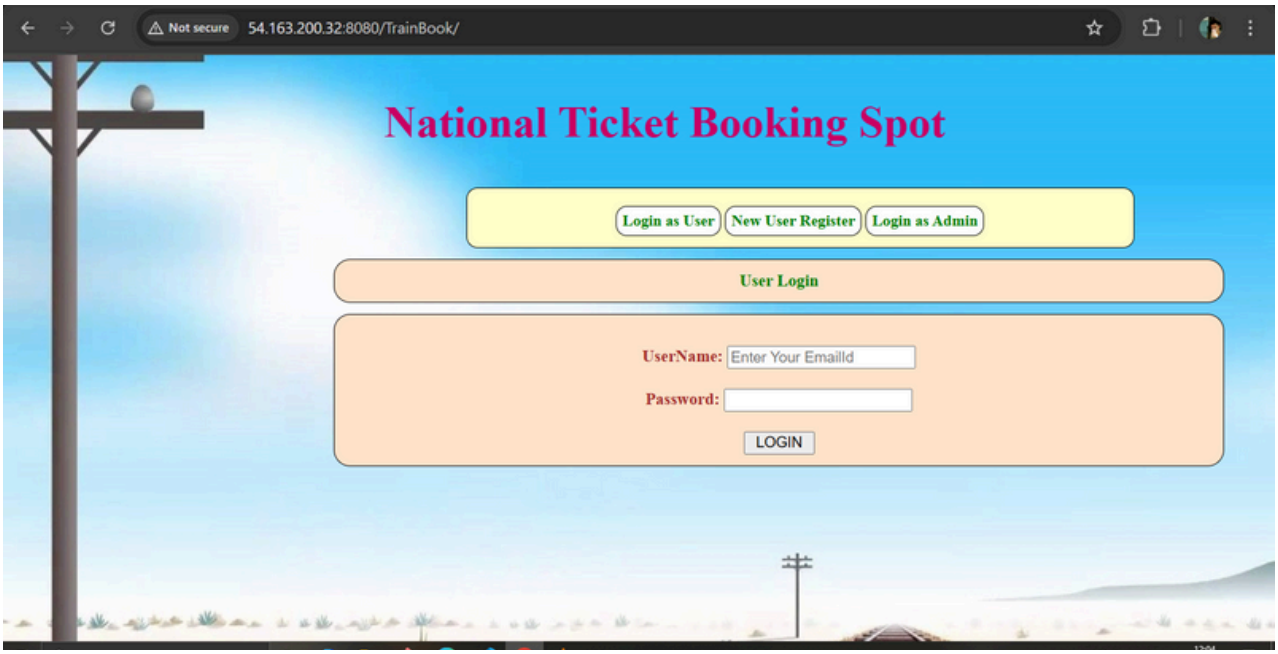
DEV Environment: Deployment ran automatically without manual approval.



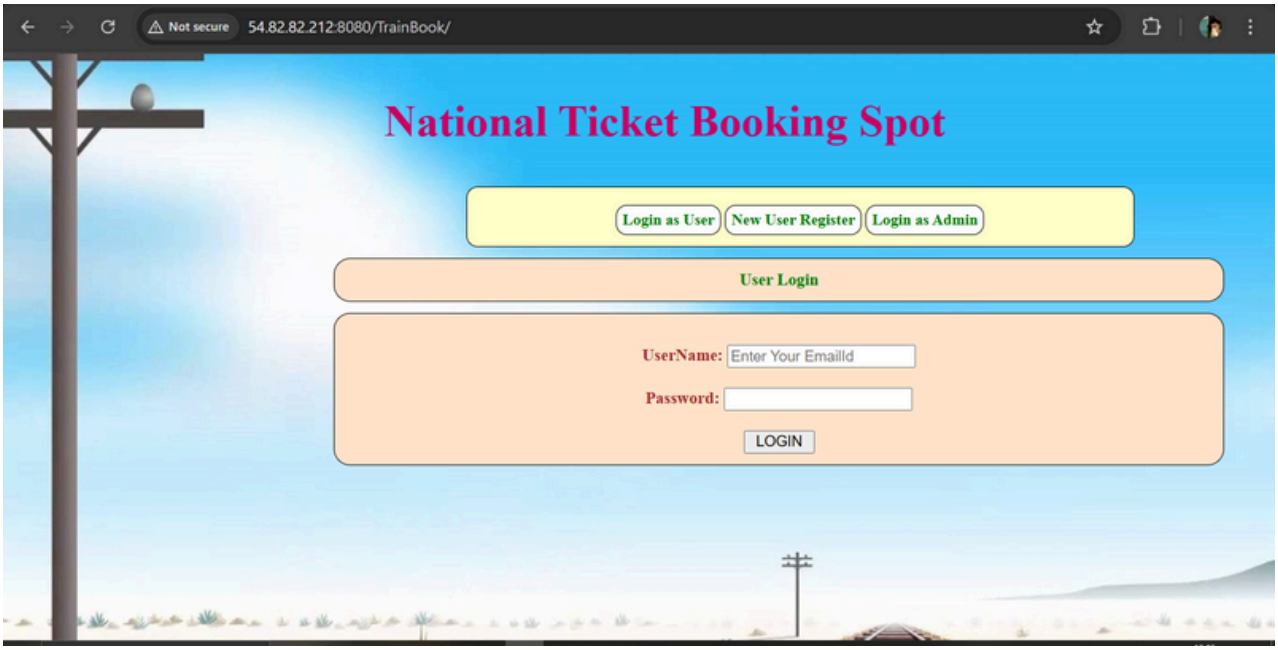
TEST Environment: Deployment required manual approval before running.



PRE-PROD Environment: Deployment required manual approval before running.



PROD Environment: Deployment required manual approval before running.



Conclusion

- The workflow successfully builds the Java application with Maven and deploys it to multiple environments on AWS EC2 Tomcat servers.
- Manual approvals add a layer of verification, preventing accidental deployments to higher environments.
- This project highlights the importance of automated CI/CD pipelines in modern software delivery and demonstrates real-time deployment control using GitHub Actions.

DEV → TEST (Approval) → PRE-PROD (Approval) → PROD (Approval)

See the Pipeline code --> GitHub